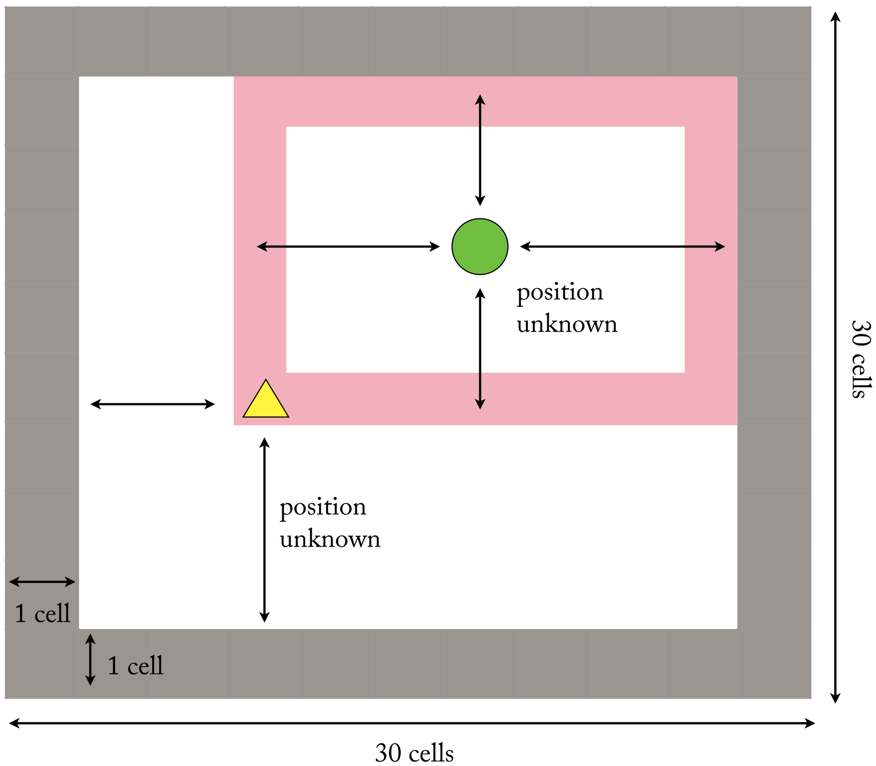
HW6. Navigation with Turbots

The Desert Ant

Goal: Search until you find food (and pass over it), then return to where you started as quickly as possible. You may choose any initial configuration for the tape.

- **Assumption 1:** the food will always appear in the NE quadrant relative to the Turbot's starting position (and not at the same position as the Turbot itself.)
- **Assumption 2:** the world is 30 X 30 cells, surrounded by a wall.
- **Constraint:** use a maximum of 20 cells on the Turing Machine tape.
- **Unknowns:** the starting position of the Turbot in the world is unknown, the position of the food within the NE quadrant of the Turbot is also unknown.



1. **Flow-chart.**

   Design a high-level flow-chart that describes your solution to the Desert Ant Problem. Be sure to describe both the design of the machine and the configuration of the tape.

2. **Turbot**

   Construct a TM that follows flow-chart in in (1) and solves the Desert Ant Problem. Use modular boxed functions when you can. Label all the major parts of your design according to their function.

   For very similar functions, it is sufficient to provide one solution, and indicate the obvious change to the other. (For example, if you have one sub-part that handles left turns, and a symmetrical one that handles right turns, you can just design one of these, and describe the other one in words.)

3. **The Life Cycle**

   Suppose that, after solving the Desert Ant Problem, your Turbot must stay at home for a fixed amount of time (say: 20 clock cycles), then return to the food, and the whole cycle must repeat again.

   Assuming the food never moves, how would you change your initial design to accommodate this challenge? How would you change the tape? Describe your approach in prose, or with the help of a flow chart, but don't design the actual Turbot.